



SMART INDIA HACKATHON 2026

Problem Statement ID: 26181

Problem Statement Title: A secure, AI-powered Personal Health Companion that delivers real-time, privacy-preserving health monitoring and early warning capabilities

Theme: MedTech / BioTech / HealthTech

PS Category: Hardware

Team Name: Omnis Cura

Team ID:

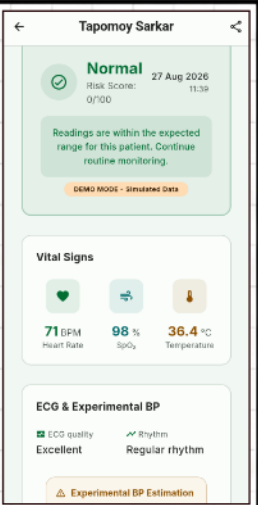
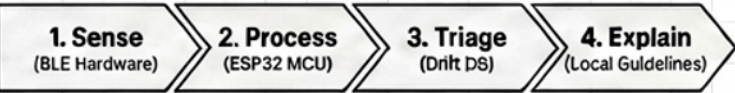
SwasthyaSetu AI

Proposed Solution

SwasthyaSetu AI is a **₹2500 Edge-AI Hardware + Flutter App Ecosystem**

Operates entirely offline (Zero-Dependency), turning raw vitals into plain-language clinical triage without requiring internet, power grids, or high medical literacy.

Detailed Explanation



How It Addresses The Problem

To protect India's vulnerable populations during recurring extreme weather and health emergencies, a robust physiological and environmental monitoring solution is required. The proposed privacy-preserving, offline-capable system will continuously track essential vital signs to provide **real-time anomaly detection, disaster-specific alerts, and emergency assistance.**

Innovation & Uniqueness

	Traditional	SwasthyaSetu AI
Infrastructure	Cloud-dependent	100% Offline (MBTiles & SMS SOS)
Medical Logic	Diagnostic Guessing (LLM hallucination risk)	Deterministic Triage with One-Way Provenance
Form Factor	Expensive / Fragile	125x125mm plus-shaped 3D printed rugged edge device

TECHNICAL APPROACH

Technologies Used

Hardware

ESP32-WROOM-32
AD8232 (ECG)
MAX30102 (SpO₂)
MLX90614 (Temp) TP4056

Software

Flutter 3.47+ SQLite (Drift)
BLE MBTiles

AI / Logic

Deterministic Rule Engine
Local Guidelines
Open-Meteo Gemini AI (Opt.)

System Architecture Flowchart

Edge Acquisition



I2C/ADC
Sensor Board
MAX30102 (SpO₂)
MLX90614 (Temp)

Edge MCU

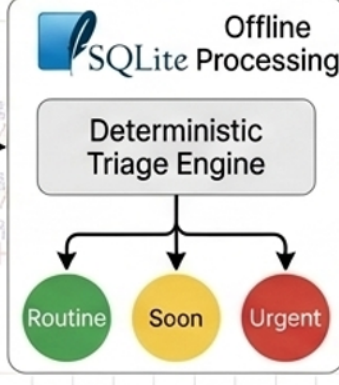


Honest Link
State BLE
Advertising

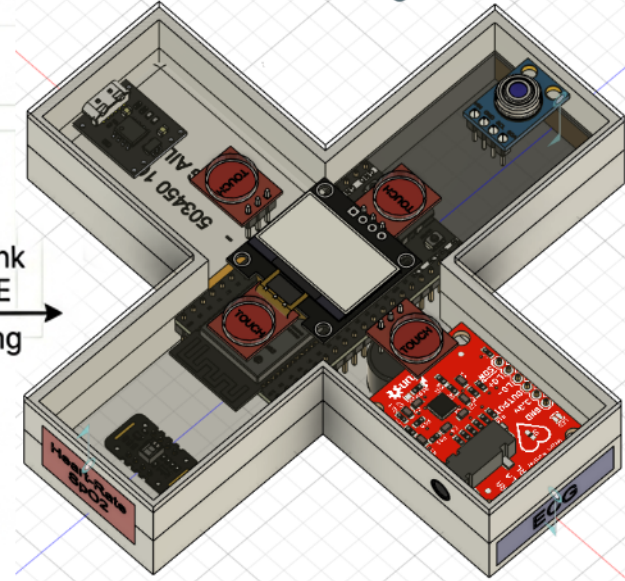
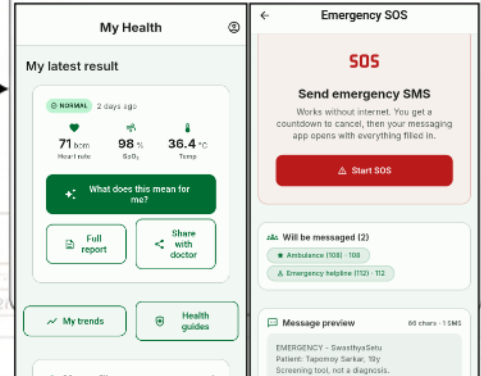
BLE

App Ingestion

Flutter App
Foreground
Service



Action Interface



FEASIBILITY AND VIABILITY

Economic Feasibility

Component (Cost)			
ESP32 DevKit (₹330)	MAX30102 (₹300)	AD8232 (₹500)	MLX90614 (₹400)
SSD1306 OLED (₹150)	1000mAh Li-ion + TP4056 (₹400)		3D Enclosure/Misc (₹420)
Total Est. Hardware Cost: ₹2500 per unit			

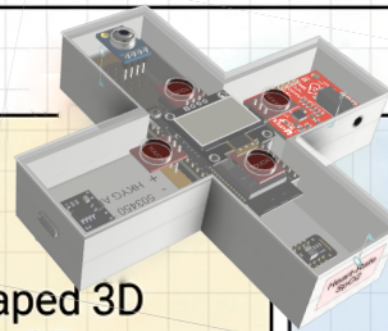
Challenges & Strategies

Risk 1: Medical Liability

Fixed
Fixed clinical deterministic rules for triage (not AI). AI is strictly isolated to explanations only.

Risk 2: Hardware Fragility

Fixed
125x125mm plus-shaped 3D printed enclosure (35x35mm center) with integrated bottom storage for band-aids and ECG pads. Reusable Li-ion battery.



Risk 3: Total Internet Blackout

Fixed
Bundled OpenStreetMap tiles (MBTiles) and SMS-based SOS payload parsing that opens native messaging apps offline.

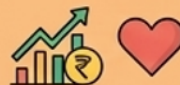
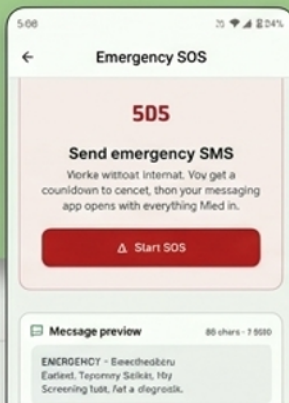
IMPACT AND BENEFITS



Target Audience (ASHA Workers)

Empowers frontline community health workers to make clinical-grade escalation decisions.

Removes the burden of interpretation; vitals instantly convert to clear action steps (e.g., "Go immediately, whatever the score says, if...").



Socio-Economic Impact

Converts a one-time ₹2500 CapEx into thousands of saved hospitalizations.

Early, decentralized detection of Sepsis, Hypoxia, and Arrhythmias drastically reduces late-stage ICU admissions and rural mortality rates.



Environmental & Scalability

E-waste reduction via reusable TP4056 rechargeable Li-ion ecosystem and standard disposable Ag/AgCl snap electrodes.

Zero-cloud-compute carbon footprint for routine screenings; data only syncs via consent-gated uploads when bandwidth permits.



RESEARCH AND REFERENCES

1. Clinical Frameworks & Triage Guidelines

- **Sepsis Detection:** WHO / ICMR Sepsis Guidelines utilized for fever, fast pulse, and low oxygen deterministic thresholding.
- **Childhood Illness:** WHO IMCI (Integrated Management of Childhood Illness) standards integrated for pediatric oxygen saturation and resting pulse baselines.
[Citation](#)
- Fraser, H. S. F., Cohan, G., Kochler, C., Anderson, J., Lawrence, A., Pateña, J., Bacher, I., & Ranney, M. L. (2022). Evaluation of Diagnostic and Triage Accuracy and Usability of a Symptom Checker in an Emergency Department: Observational Study. JMIR mHealth and uHealth, 10(9), e38364.

2. Technical Architecture & Hardware

- **Mapping:** ODBL OpenStreetMap architecture for offline .mbtiles generation.
- **Firmware:** ESP-IDF documentation for Bluetooth Low Energy UUID service structuring and I2C/ADC sensor polling.
[Citation](#)
- Jebakumar, S. S., Hemalatha, R. J., Malhotra, S., Shivapanchakshari, T. G., Lokesh, K., & Anand, D. D. (2025). IoT-Enabled Hemodynamic Surveillance System: AD8232 Bioelectric Signal Processing with ESP32. arXiv preprint arXiv:2505.18173.

3. Open-Source Verification

- **Repository:** GitHub - <https://github.com/krishnendukoley2007-arch/swasthyasetu-ai>
- **Engineering Rigor:** Software backed by unit/widget tests ensuring deterministic rule accuracy and offline map honesty. Validated UI overflow handling for maximum accessibility.

